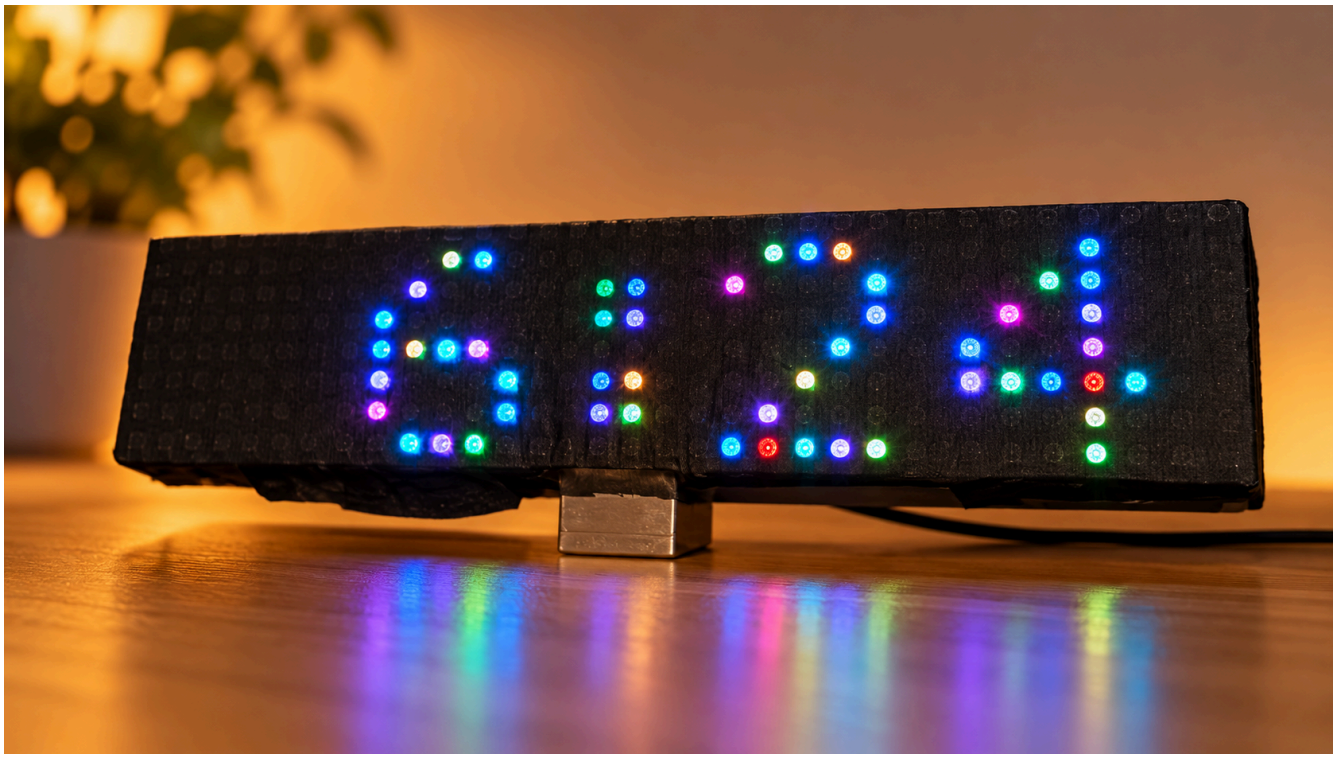
ESP Siesta Clock with Matrix Display

Technical Reference - English



A clever, super-dark clock that respects your sleep.

1. System overview

The final firmware is an ESP8266 clock controller integrating a 256-pixel WS2812 display, DS3231 RTC, SSD1306 OLED, direct NEC IR decoder, EEPROM configuration, bounded WiFi/NTP sessions and a public-holiday cache. The The repository contains the final working firmware used by the device.

2. Architecture

Subsystem	Role
ESP8266	Main state machine, display, editor, networking and IR decode
DS3231	Continuous time source and fallback
WS2812 32 x 8	Primary time display
SSD1306 128 x 64	Configuration/status UI
NEC IR	Primary user input
GPIO0 button	Boot-time WiFiManager portal trigger
EEPROM 128 bytes	Persistent settings and holiday cache
NTP	Periodic correction of RTC
Nager.Date endpoint	Nationwide public-holiday source when Public rule is enabled

3. Pin assignment

Signal	GPIO
WS2812 data	13
IR receiver	14
Config button	0
OLED/RTC SDA	4
OLED/RTC SCL	5

4. Display geometry

Matrix constants: 32 columns x 8 rows, 256 WS2812 pixels. Digit X positions are 2, 8, 19 and 25; DIGIT_Y is 1. The breathing separator uses columns 15 and 16 with lit sections on rows 2-3 and 5-6 and a black row 4. Brightness range is 1-30; fresh-install day brightness is 6.

5. Timekeeping and WiFi lifecycle

The DS3231 is the continuous time source. On normal boot the firmware does not call WiFiManager.autoConnect(). Instead it attempts saved credentials for a bounded 20-second window. If connected, it synchronizes NTP, updates the RTC, optionally refreshes public-holiday data, then calls the WiFi shutdown path. Holding GPIO0 low at boot explicitly enters WiFiManager configuration.

The default AUTOSYNC interval is 24 hours, defined by DEFAULT_SYNC_HOURS. The interval is user-selectable from 1 to 255 hours.

6. Country and timezone model

Country and Time Zone are intentionally separate. The firmware exposes 249 country entries for holiday data and 63 compact timezone rules for civil-time behavior. Greece is index 0. Country selection never automatically changes timezone.

7. Holidays and weekend engine

Holidays has a master enable plus three independent date rules: nationwide Public Holidays, Saturdays and Sundays. A matching rule changes only the effective Night Mode end time to holidayOverrideHour:holidayOverrideMinute. The Night Mode start time and all other clock behavior are unchanged. Fresh-install override time is 08:00.

Public holiday data is cached as a 366-day bitmap occupying 46 bytes in EEPROM. Weekend-only operation is local and requires no Internet access. When Public Holidays is disabled, the holiday engine does not need to refresh online public-holiday data for Saturday/Sunday matching.

8. Night Mode priority

Night Mode is the brightness authority. checkBrightnessMode() computes the effective end time, including a holiday/weekend override when applicable, then enforces BRIGHTNESS_MIN. Purple editor override may change

color but is not allowed to raise brightness. This is the implementation behind the project message: “A clever, super-dark clock that respects your sleep.”

9. Seasonal Siesta engine

Season	Day	Night	Calendar
Summer	15:00-17:30	23:00-07:00	01/04-30/09
Winter	15:30-17:30	22:00-07:30	01/10-31/03

The editor copies the saved seasonal structures into temporary edit structures and commits only after all eight steps validate. This prevents partially edited schedules from being written to EEPROM.

10. OLED editor

The EditorPage enum contains: MAIN, COUNTRY, TIMEZONE, TIME, HOUR_FORMAT, FONT, SIESTA, NIGHT, HOLIDAYS, HOLIDAY_OPTIONS, DIMMER, SYNC, MONO_COLOR, MULTI_COLOR and LANGUAGE. The main menu presents 14 user-facing entries. English and Greek labels are implemented directly in firmware.

11. Fonts and color modes

Six matrix font styles are available. The DayColorMode enum provides MONO, SOLID, COLORFUL, FUNNY and DYNAMIC behavior. Mono mode uses a 16-color palette. Orange is reserved for Siesta indication and intentionally excluded from the user mono palette.

12. EEPROM map

Address	Content
0-3	Night Mode start/end
4	Day brightness
5	Legacy timezone byte
6-9	Reserved legacy Siesta bytes
10	Mono color
11	AUTOSYNC hours
12-14	Unused
15-16	Day color mode + magic
17-41	Seasonal Siesta settings + magic
42-44	12/24, font, display magic
45-46	OLED language + magic
47	Country index
48	Holidays master
49-50	Holiday override HH:MM
51	Holiday settings magic
52-55	Holiday cache metadata
56-101	366-day holiday bitmap
102-104	Extended timezone index + magic
105	Holiday rule bit mask: Public/Saturday/Sunday
106	Holiday-rule magic
107-127	Unused

13. Dependencies

- ESP8266WiFi.h
- ESP8266HTTPClient.h
- WiFiClientSecureBearSSL.h
- TZ.h
- WiFiManager.h
- NTPClient.h
- WiFiUdp.h
- Adafruit_NeoPixel.h
- EEPROM.h
- U8g2lib.h
- RTClib.h

14. Network and security note

The firmware uses WiFi only in bounded sessions. Public-holiday HTTPS retrieval uses the ESP8266 BearSSL client. In this build, the holiday download path uses an insecure certificate-validation mode for compatibility; transport remains encrypted, but the firmware does not authenticate the remote server certificate. This is an implementation fact to consider if the project is adapted for a higher-security environment.

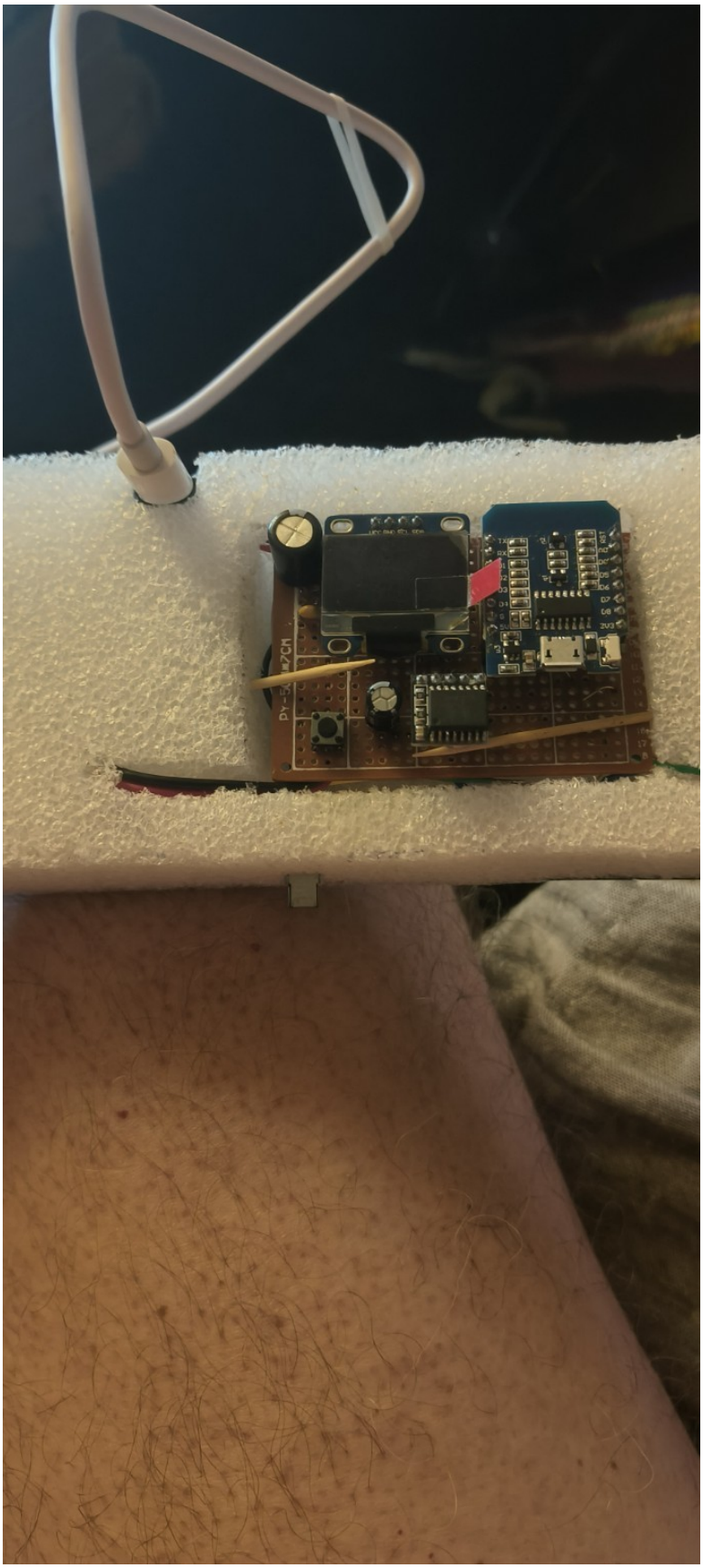
15. Source identity

Publication filename: firmware/ESP_Siesta_Clock_with_Matrix_Display.ino
Lines: 7,979

Bytes: 177,326

SHA-256: 8e842bcc9da1b2affbb206ff4803113f7d8c407f40a1ed76053045c1f0a75b74

The publication source uses the verified final firmware logic and concise operational comments.



Rear electronics assembly in the photographed prototype.